

The most accurate atomic clocks will not lose or gain more than 1 second in 15 billion years.

FREQUENCY AND TIME

The definition of a second has been based on this frequency since 1968.

9,192,631,770
OSCILLATIONS

RADIO WAVE SIGNAL
SENT AT 9,192,631,770 HZ

OSCILLATOR

FEEDBACK TO
OSCILLATOR

A second magnet filters out low-energy cesium atoms before a detector counts the number of atoms. If the detector counts enough high-energy-state atoms, then the oscillator is at the right frequency. If the number of high-energy atoms is too low, then the oscillator needs adjusting to the correct frequency.

MAGNET

MAGNET

DETECTOR

LOW-ENERGY-STATE
ATOMS REMOVED